

Network Cable Types and Tools

Task 1: Ethernet Cable vs Fiber Optic Cable

Feature	Ethernet Cable	Fiber Optic Cable
Transmission Medium	Copper wires	Glass or plastic optical fibers
Speed	Up to 10 Gbps (depending on cable category)	Up to 100 Gbps and beyond
Distance	Up to 100 meters	Several kilometers without significant signal loss

Real-World Examples

- **Ethernet Cable:** Connecting a home Wi-Fi router to a desktop PC or gaming console.
- **Fiber Optic Cable:** Connecting buildings on a college campus or linking servers in a data center.

Task 2: Straight-Through Ethernet Cable (T568B Standard)

RJ45 Connector (Front View)

Pin	Wire Color
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

T568B Wiring Order

Pin	Wire Color
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- | | |
|---|--------------|
| 1 | White/Orange |
| 2 | Orange |
| 3 | White/Green |
| 4 | Blue |
| 5 | White/Blue |
| 6 | Green |
| 7 | White/Brown |
| 8 | Brown |

Note: A straight-through cable uses the **same T568B wiring order on both ends**.

Task 3: Steps to Attach an RJ45 Connector to a CAT6 Cable

Materials Required

- CAT6 Ethernet cable
- RJ45 connector
- RJ45 crimping tool
- Cable stripper
- Cable tester

Procedure

1. Measure the required cable length.
2. Strip approximately 2–3 cm of the outer cable jacket.
3. Untwist the wire pairs carefully.
4. Arrange the wires in the **T568B color order**.
5. Straighten and trim the wires so they are all the same length.
6. Insert the wires fully into the RJ45 connector, ensuring each wire reaches the end.
7. Place the connector into the RJ45 crimping tool.
8. Squeeze the crimper firmly to secure the connector.
9. Repeat the same wiring order on the other end for a straight-through cable.
10. Test the finished cable using a network cable tester.

Task 4: Gaming Café Network

For a gaming café with **10 PCs**, I would choose **CAT6 Ethernet cables** because they provide high-speed, reliable connections at a lower cost than fiber optics for short distances. A **RJ45 crimper** would be used to attach RJ45 connectors to the CAT6 cables. Fiber optic cables are better suited for long-distance links between buildings or server rooms, whereas Ethernet is ideal for connecting computers within the same room. This setup offers excellent performance for LAN gaming while staying within budget.

Conclusion

Ethernet cables are cost-effective and ideal for home and office networks, while fiber optic cables are preferred for high-speed, long-distance communication. Proper use of the T568B wiring standard and an RJ45 crimping tool ensures reliable Ethernet cable connections.